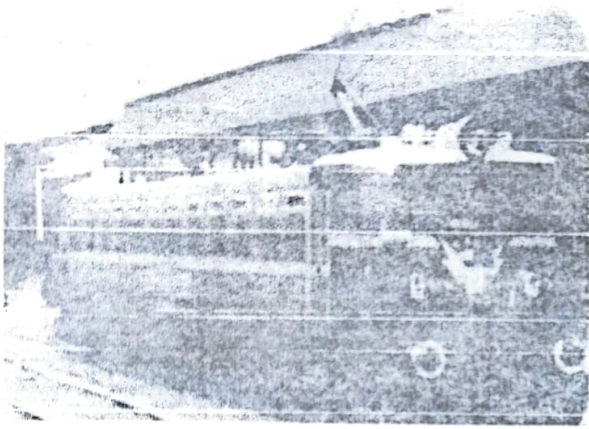


DECCAN WESTERN RAILWAY



इलेक्ट्रिक लोको शेड वलसाड

बोगी सेक्शन AOH/IOH/TOH चेक शीट

LOCO NO. : 32649 . WAG. 9.
SCHEDULE : TOH.
DATE OF COMMENCEMENT : 27-03-23
DATE OF COMPLETION : 12-04-23

SSE/M4712/BL

ELECTRIC LOCO SHED, VALSAD
WAG-9 LOCO (TOH) CHECK SHEET

Loco No. 32649

Schedule TOH

Date _____

SN	Schedule Activity	Standard	Actual/Action taken				
1	BOGIE FRAME OVERHAULING (RDSO Letter No. EL/3.1.35/16 Dated 30.03.2007)						
1	Remove all brake rigging items from the bogie frame before overhaul of bogie frame.(IOH)	Removed	Removed				
2	Clean the bogie frame by scrapping.	Cleaned	cleaned				
3	Make a through visual inspection of bogie frame for cracks, damage and evidence of fatigue.	No crack/ No damage	OK				
4	RDPT of TBU hanger bracket tube at weld for any crack. (SMI/0251)WAG9H	No crack	N/A				
5	RDPT of the brake hanger and lever to detect any crack.	No crack	OK				
6	Critically check the bogie motor support (lug) for any crack by MPT.	Checked	OK				
7	Check the brake rigging for loose, damage or worm out parts. Rectify as required. (TOH)	Checked/ Rectified	checked				
8	Replace all pins, bushes and fasteners of brake rigging with new one. (IOH)	Replaced	Replaced				
9	Replaced the brake blocks with new one.	Replaced	Replaced				
10	Check the brake block key security. Ensure intactness of R-clip and secure with cable tie.	Ensured	N/A as Conv. B/R				
11	Provide safety slings on all TBU + PBU location. (SMI/0244) WAG9H	Provided	Provide d				
12	Check rubbing of brake hanger with bogie frame.	No rubbing	No rubbing				
	Minimum gap to be maintained between bogie frame & hanger. (SMI/251) WAG9H N/A as Conv. B/R	As per table below	As per table below				
	Standard	1	2	3	4	5	6
	Gap 13mm	✓	✓	✓	✓	✓	✓
13	Replace the lateral and vertical stops.	Replaced	Replaced				
14	Modification of top bracket for secondary vertical dampers of WAG9 loco to prevent their failure MP.MOD.VL.02.07.09 (Rev.-00)	Replaced	Modified				
15a	Ensure 100% availability of slack adjuster. (SMI/0251) WAG9H	Available	Available				
b	Check length of spring after tightness N/A as Conv. B/R						
	Standard	1	2	3	4	5	6
	70 mm	✓	✓	✓	✓	✓	✓
c	Check gap between lug and slack adjuster.						
	Standard	1	2	3	4	5	6
	7.5 mm	✓	✓	✓	✓	✓	✓

Name of Staff

Signature of SE/JE


 Signature of SSE

ELECTRIC LOCO SHED, VALSAD
WAG-9 LOCO (TOH) CHECK SHEET

Loco No. 32649

Schedule TOH

Date _____

SN	Schedule Activity	Standard	Actual/Action taken				
16	Replacement of all sphere blocks.	Replaced					
17	Check the earthing shunts provided on axle box, bogie & body.	Provided	Provided				
18	Maintain the vertical & lateral clearance (Primary & secondary) as per RDSO/TC/082	As per table below	As per table below				
A. Primary vertical clearance (on one bogie)							
STD		Actual values					
	1	2	3	4	5	6	
27 to 35	35	31	32	32	31	30	
B. Primary lateral clearance (on one bogie)				Actual values			
STD		1	2	3	4	5	6
15 to 22	18	19	18	17	20	20	
AVG							
C. Secondary vertical clearance (on one bogie)				Actual values			
STD		1		2			
32 to 40		37		38			
D. Secondary lateral clearance (on one bogie)				Actual values			
STD		1		2			
45 to 55		50		42			
AVG							
II	OVERHAUL OF TBU/PBU WAG9H						
1	MPT of brake lever as per SMI 251	Done		} N/A as			
2	Replacement of brake lever if found cracked/deformed	Replaced		} Conv. B/R			
3	Replacement of threaded bolts and F.S all steel lock nuts.	Replaced					

Name of Staff

Signature of SE/JE

Signature of SSE

ELECTRIC LOCO SHED, VALSAD
WAG-9 LOCO (TOH) CHECK SHEET

Loco No. 32649

Schedule TOH

Date 28/3/23

SN	Schedule Activity	Standard	Actual/Action taken
III	SUSPENSION SPRING		
a	Examine the primary and secondary springs for signs of damage cracks etc.	No damage/ No cracked	No damage
b	Measure primary and secondary springs characteristic/parameter.	As per table below	measured

Check Sheet for WAG-9H Secondary Outer Spring
(CLW Drg. No. 1209.01.015.010 Alt.01) IB 011-00842

SN	Test Parameter	STD	Actual			
			1/5	2/6	3/7	4/8
1	Free height	733.2 mm	736/733	744/733	736/733	736/733
2	Working height	552.0 - 609.5	588/583	587/584	587/585	586/586
3	Spring working load	9871 Kg	—	9871 kg	—	—
4	The end tip should not bite the effective coil	No bite	—	No bite	—	—
5	None of coil shall be in contact with adjacent coil at working height	No touching	—	No touching	—	—
6	Band 1 positive tolerance Band 2 negative tolerance	575.0 - 609.5 mm 552.0 - 574.0 mm	—	within limit	—	—
7	Spring of same bend should be used in same bogie	Followed	—	Followed	—	—

Check Sheet for WAG-9H Secondary Inner Spring
(CLW Drg. No. 1209.01.015.010 Alt.01) IB 011-00842

SN	Test Parameter	STD	Actual			
			1/5	2/6	3/7	4/8
1	Free height	491 mm	496/495	496/495	494/493	493/494
2	Working height	388.8 - 429.5 mm	407/406	407/404	406/404	406/404
3	Spring working load	1499 Kg	—	1499 kg	—	—
4	The end tip should not bite the effective coil	No bite	—	No bite	—	—
5	None of coil shall be in contact with adjacent coil at working height	No touching	—	No touching	—	—
6	Band 1 positive tolerance Band 2 negative tolerance	405.0 - 429.3 mm 388.8 - 404.0 mm	—	within limit	—	—
7	Spring of same bend should be used in same bogie	Followed	—	Followed	—	—

Satyam (App)
Dilip (App)
Abhimanyu (App)

Name of Staff

Signature of SE/JE


Signature of SSE

ELECTRIC LOCO SHED, VALSAD
WAG-9 LOCO (TOH) CHECK SHEET

Loco No. 32649

Schedule TOH

Date _____

Check Sheet for WAG-9H Primary Outer Spring
(Ref. : CLW Drg. No. IB 011.00843)

SN	Test Parameter	STD	Actual											
			1	2	3	4	5	6	7	8	9	10	11	12
1	Free height	238.8 mm	<u> </u> Bogie not removed											
2	Working height	184.3-203.5 mm												
3	Spring working load	4140 Kg	bogie not started, Visually checked OK											
4	The end tip should not bite the effective coil	No bite												
5	None of coil shall be in contact with adjacent coil at working height	No touching												
6	Band 1 positive tolerance Band 2 negative tolerance	192.0-203.5 mm 184.3-191.0 mm												
7	Spring of same bend should be used in same bogie	Followed												

Bogie not removed

bogie not tested, Visually checked OK

Name of Staff

Signature of SE/JE

Signature of SSE

ELECTRIC LOCO SHED, VALSAD
WAG-9 LOCO (TOH) CHECK SHEET

Loco No. 39645

Schedule TOH

Date _____

Check Sheet for WAG-9H Primary Inner Spring
 [Ref. : CLW Drg. No. IB 011.00843]

SN	Test Parameter	STD	Actual											
			1	2	3	4	5	6	7	8	9	10	11	12
1	Free height	240.2 mm												
2	Working height	180.5-199.3 mm												
3	Spring working load	500 Kg												
4	The end tip should not bite the effective coil	No bite												
5	None of coil shall be in contact with adjacent coil at working height	No touching												
6	Band 1 positive tolerance Band 2 negative tolerance	188.0- 199.3 mm 180.5-187.0 mm												
7	Spring of same bend should be used in same bogie	Followed												

Bogie not removed

hook not tested, Visually checked OK

→

Name of Staff

Signature of SE/JE

Signature of XSE

ELECTRIC LOCO SHED, VALSAD
WAG-9 LOCO (TOH) CHECK SHEET

Loco No. 32649

Schedule TOH

Date _____

SN	Schedule Activity	Standard	Actual/Action taken					
III c	Ensure pairing of springs as per band.	Ensured	Ensured					
IV	DAMPERS (RDSO Letter No. EL/3.1.35/16 Dated 30.03.2007)							
a	Replace all dampers	Replaced	—————					
b	Replace all FS all steel lock nuts opened for lifting of locos at mentioned location. A. Secondary dampers (Yaw, Vertical, Horizontal) B. Primary dampers top side	Replaced on conditional basis	Replaced on conditional basis					
c	Check the alignment of top bracket foundation of damper with bogie frame	Aligned	checked OK					
V	SPHERI BLOCS (RDSO Letter No. EL/3.1.35/16 Dated 30.03.2007)							
a	Replace all the Spheri bloc of axle guide, and traction motor torque arm	Replaced	All Traction Arm S/blocks new provided					
b	Check intactness of axle guide bolt	Intact	Intact					
c	Ensure tightness by specified torque							
	Bolt type	Std torque	Actual					
			1	2	3	4	5	6
	Necked down	670 Nm	————— Tightness	checked	OK	—————		
	M-24×300	665 Nm	————— Tightness	checked	OK	—————		

N. K. Panchal
Eshwar

Name of Staff

Signature of SE/JE

Signature of SSE

ELECTRIC LOCO SHED, VALSAD
WAG-9 LOCO (TOH) CHECK SHEET

Loco No. 32649

Schedule TOH

Date 31/4/23

SN	Schedule Activity	Standard	Actual/Action taken
VII	MSU (RDSO Letter No. EL/3.1.35/16 Dated 30.03.2007)		
1	Greasing to be done in motor suspension unit (MSU) at NDE side	100 gm	Regreasing done
2	Check and tighten all fasteners of NDE side	Checked	checked & tightened
VIII	AXLE (RDSO Letter No. EL/3.1.35/16 Dated 30.03.2007)		
1	Disconnect the electrical cable from the earth return unit. Remove the earth return unit from the axle. Examine the bearings and rear sealing position for signs of grease leakage or discoloration.	Done	done
2	Test the axle for surface and sub-surface defects using a non-destructive technique (Ultrasonic flaw detector). Replace the axle if cracked, defective or damaged.	Good condition	Axle UST done
IX	AXLE BOX (RDSO Letter No. EL/3.1.35/16 Dated 30.03.2007 and RDSO's SMI/0246)		
1	Visually inspect the axle boxes for wear or damage. Pay particular attention to the guide rod mounts and limit stops. Replace the axle box if defective.	Good condition	Visually checked OK
2	Open the axle journal box and inspect the condition of lubricating grease, inner races, rollers and cages. Sample the lubricants and analyze for contamination/ metal content.	Done	checked OK
3	UST of axle racer.	Done	Axle UST done
4	Replace the bearing along with its accessories if worn or damage. (NEI: RB5080, SKF:CRU150)	Replaced	Servicable
5	Replace "O" ring of rear cover. (IOH)	Replaced	Replaced
6	Ensure tightness of rear cover bolts (M 16x45) by specified torque. (IOH)	Torque 100Nm	Tightened
7	Replace the grease of bearing	590 gm	Replaced
8	Tighten the end plate socket head caps screw (M-20x50) with specified torque.	Torque 33mkg	Tightened done
9	Measure the wear of carbon brush and see it does not go below the grooved line and replaces it if required.	Measured/ Replaced	Servicable
10	Check the intactness of the earth return path on each wheel set.	Done	Intact
11	Replace the axle box front cover 'O' ring.	Replaced	Replaced
12	Ensure tightness of front cover bolts (M 16x45) by specified torque.	Torque 100Nm	Tightness done

Raman
Shantilal
Kantv

Name of Staff

Signature of SE/JE

Signature of SSE

ELECTRIC LOCO SHED, VALSAD
WAG-9 LOCO (TOH) CHECK SHEET

Loco No. 22649

Schedule TOH

Date _____

SN	Test	Standard		Wheel Set		
				1	2	3
1	Sr. No. (Axle No.)			TG 28-47	TG 71-79	TG 62-18
2	UST of Axle (TC-47)			Done	Done	Done
3	Check wheel thickness	133 + 1 - 0	PE	133.8	134.2	134.5
			CE	133.5	134.3	134.5
4	Check wheel diameter	New : 1097.0 Service Limit : 1016.0	PE	1045	1050	1056.50
			CE	1045	1050	1056.00
5	Check root wear	6.0 mm	PE	4.0	0.0	1.0
			CE	2.0	0.0	3.5
6	Check flange wear	3.0 mm	PE	1.0	0.0	0.5
			CE	0.5	0.0	1.0
7	Check trade wear	6.5 mm	PE	2.5	0.0	3.0
			CE	1.5	0.0	1.0
8	Check inside face wheel distance (According to MPM1.BD.02.16.01 (Rev-01) Dt:31/1209)	1596 + 1.5 mm - 0.5 mm Service limit : + 3.0 (1599)		1596.10	1596.0	1597.37
9	<u>Check 'K' value:</u> 107:21 T New limit (317.09 - 317.17)mm - 15 TEETH Condemning limit : 316.494mm			317.10	317.11	317.09

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 Signature of SSE

ELECTRIC LOCO SHED, VALSAD
WAG-9 LOCO (TOH) CHECK SHEET

Loco No. 32649

Schedule TOH

Date 31/3/23
03/4/23

Check Sheet for Axle Box Overhauling of WAG-9 Loco
(Ref. : CLW Drg. No. 1209.01.011.088 Alt.01)

SN	Test Parameter	STD	Actual					
			1	2	3	4	5	6
1	Axle box housing	250 + 0.02 mm						
	(a) Bore dia	- 0.00 mm						
	(b) Bore length	220 + 0.10 mm						
		- 0.10 mm						
2	Bearing cage condition	Good condition	—	Good	—	—	—	—
3	Bearing roller condition	Good condition	—	Good	—	—	—	—
4	Condition of rear cover	Good condition	—	Good	—	—	—	—

Check Sheet for Axle Box Overhauling of WAG-9 Loco
(Ref. : CLW Drg. No. 1209.01.011.088 Alt.01)

SN	Test Parameter	STD	Actual					
			1	2	3	4	5	6
1	Abutment ring seat dia.	Std. 170 + 0.146 mm - 0.186 mm	✓	✓	✓	✓	✓	✓
2	Abutment ring inner dia.	Std. 170 + 0.106 mm - 0.043 mm	✓	✓	✓	✓	✓	✓
3	Abutment ring interference	0.04 to 0.143 mm	✓	0.08 0.07	✓	✓	✓	✓
4	Distance from axle end to abutment ring end	(188.0 to 189.0)mm	✓	✓	✓	✓	✓	✓
5	Racer seat dia.	150.0 + 0.043 mm - 0.068 mm	✓	✓	✓	✓	✓	✓
6	Racer inner dia.	(150.0 - 0.025)mm	✓	✓	✓	✓	✓	✓
7	Racer interference	(0.068 to 0.093)mm	✓	✓	✓	✓	✓	✓
8	Racer OD before fitment	174.0mm	✓	✓	✓	✓	✓	✓
9	Racer OD after fitment	—	✓	✓	✓	✓	✓	✓
10	Distance from axle end to racer end	(22.0 to 23.0)mm	✓	✓	✓	✓	✓	✓
11	Racer make	NBC/SKF/FAG	NBC NBC	SKF SKF	—	NBC NBC	—	—

Dhiru Babulal
Ramesh, Pravin (App)
Jayesh
Ramesh

Name of Staff

Signature of SE/JE

Signature of SE/JE

ELECTRIC LOCO SHED, VALSAD
WAG-9 LOCO (TOH) CHECK SHEET

Loco No. 32649

Schedule TOH

Date 27/3/23

SN	Schedule Activity	Standard	Actual/Action taken
X	GEAR CASE/ Transmission (RDSO Letter No. EL/3.1.35/16 Dated 30.03.2007)		
1	Open gear case and remove gear oil & cleaning of gear case	Cleaned	<i>Cleaned</i>
2	Check breakage of bolts	Checked	<i>checked</i>
3	Open oil gauge assembly and replace rubber item and oil gauge glass	Opened Replaced	<i>opened & Replaced</i>
4	Ensure tightness of magnetic sealing plug & oil filling cap. Apply RTV	Ensured Applied	<i>Ensured Applied</i>
5	Verify the integrity of gauge glass protection cover	Done	<i>Done</i>
6	Refill 06 litres new servo syn oil(RR460) in each gear case (RDSO/TC/034 Rev-04)	Filled	<i>Filled</i>
7	Apply RTV at all joints marching faces	Applied	<i>Applied</i>
8	Coupled with following bolts with specified torque given below Hex head screw-16×200 mm Socket head screw- 12×160 mm Socket head cap screw- 12×80 mm Hex head screw- 12×30 mm Hex head screw- 12×70 mm	300(10.9)Nm 205(8.8)Nm 80Nm 80Nm 80Nm	<i>Specified torque applied</i>
9	Replace all FS all steel lock nuts opened for lifting for locos at torque arm	Replaced	<i>Replaced</i>

*Harish chaudhary
Abhimanyu (A/R)*

Name of Staff

Signature of SE/JE

(Signature)
Signature of SSE

ELECTRIC LOCO SHED, VALSAD
WAG-9 LOCO (TOH) CHECK SHEET

Loco No. 32649

Schedule TOH

Date _____

SN	Schedule Activity	Standard	Actual/Action taken
XIII	TRACTION LINK (SMI/0259)		
1	Replace all FS all steel lock nuts opened for lifting of locos at traction link. (SMI/0259)	Replaced	Replaced
2	Check traction rod flange and housing flange against crack by RDPT. (SMI/0259) Traction rod is to be de-bolted completely.	Checked	checked
3	Check centering/ alignment of traction link housing.		checked
4	Change Aclathen ring. (SMI/0259)	Changed	—
5	Replace flange bolts and nuts of traction bar of size & tightness by specified torque. (SMI/0259)		tightened done
6	i) M 20 × 65 Use Inter lock washers (Disc lock/ Nord lock) with modified bolt. (SMI/0259)	26.0 Kgm Used	Used
7	Use F.S all steel lock nuts with modified bolt M20×65. (SMI/0259) This should not be reused.	Used	Used
8	Ensure the intactness of safety cables, pins and R-clips	Ensured	Ensured
9	Check intactness/ security of inner & outer retaining ring, bolt and plate. Ensure tightness at required torque. (80 Nm for outer ring/ 60 Nm for inner ring)	Ensured	Ensured
10	10.9 property class bolts to be replaced with 8.8 property class bolts.	Replaced	Replaced
XIV	TRACTION MOTOR		
i)	Check the condition of traction motor sphere bloc. Replace if worn out, cracked, damaged or showing signs of deterioration to either the rubber or the metal.	Checked	TM 4,5,6 s/block new provided
ii)	Check the suspension tube bearing grease (NDE end) and do the greasing. Grease 100 gm/tube	Checked/ Greasing Done	checked & Greasing done
XV	Primary earthing cable and bracket		
1	Disassemble the locomotive primary earthing system. Inspect and repair, if required, the earthing cable and bracket, earth contact plate, earth brushes, earth brush springs, earth cable glands etc. at the axle box, as prescribed.	Checked/ Replaced	checked
2	Check the security of the primary earth fasteners. Tighten the fasteners if necessary	Checked	checked

TM 4,5,6 T/Arm S/block - N. K. Panchal

Name of Staff

Signature of SE/JE

Signature of SSE

ELECTRIC LOCO SHED, VALSAD
WAG-9 LOCO (TOH) CHECK SHEET

Loco No. 32649

Schedule TOH

Date _____

XV. List of must change items of WAG-9 Loco in IOH (Per loco)

SN	Items to be changed	Std. Qty	Actual done	Remark
1	Sphere bloc (Axle guide + Torque arm)	36 Nos.		Done
2	Hardware & F.S All steel lock nuts	All	All	
3	'O' Ring of Axle box	24 Nos.		Done
4	'O' Ring of Gear case	12 Nos.		Done
5	Gasket of breather & spy glass overhauling	24 Nos.		Done
6	Traction link hardware (Bolts & Lock nuts)	24 Set		Done
7	Lateral stop (Lateral & Vertical)	04 Nos.		Done
8	Heli coils	01 Set		Done
9	Earth return brush	04 Set		Done
10	Aclathen ring	04 Nos.		Done
11	Damper	01 Set		Done

XVI. Chemical test to be carried out by C & M for WAG-9 Loco

SN	Components	Test	Actual	Remark
1	Main gear	DPT	-	DPT Don
2	Bogie lug bracket	DPT	-	DPT Don
3	Traction bar	MPT	-	MPT Don
4	Housing	MPT	-	MPT Don
5	Racer	UST	-	UST Don
6	Brake hanger, lever, tube	DPT	-	DPT Don

PRIMARY AND SECONDARY SUSPENSION LATERAL AND VERTICAL CLEARANCE

Loco No. :

Type of inspection :

Date :

Limits of Clearance	Wheel No./ Axle Box No.											
	1	2	3	4	5	6	7	8	9	10	11	12
Primary vertical -27 to 35												
Primary lateral -15 to 22												
Secondary vertical -32 to 40												
Secondary lateral -45 to 55												

Name of Staff

Signature of SE/JE

Signature of SSE

ELECTRIC LOCO SHED, VALSAD
WAG-9 LOCO (TOH) CHECK SHEET

Loco No. 32649

Schedule TOH

Date _____

SN	Description	CAB-1	CAB-2
1	CBC No. (Make & Years)	FASP 06-19	FASP 05-19
Check for any crack by MPT test of CBC assembly items.			
2	A. CBC Coupler body.	OK	OK
	B. Knuckle.	OK	OK
	C. Knuckle Pin.	OK	OK
	D. Clevis.	OK	OK
	E. Clevis Pin.	OK	OK
3	Checking of knuckle nose wear & stretch with Go-No Go gauge (Gauge No-3) New - 25.0 mm, Cond.- 9.5 mm	OK	OK
4	Checking gap between knuckle nose & coupler by gauge no 1 & 2 Gauge 1-135.0 mm Gauge 2-130.0 mm (AOH)	checked OK	checked OK
5	Check knuckle pin diameter. New - 41.2 mm, Service limit - 38.0 mm	40.00	40.00
6	Check clevis pin diameter. New - 38.0 mm, Service limit - 36.0 mm	37.00	37.00
7	Check working of CBC & Anti creep test.	Working OK	Working OK
8	Check condition & proper working of CBC operating handle.	Working OK	Working OK
9	Check for any crack and damage in yoke & draft gear.	No damage	No damage
10	Check free lateral movement of CBC shank.	OK	OK
11	Check condition of striker casting liner. New - 8.0mm, Cond.- 2.0 mm	7.0	7.0
12	Check condition of shank wear liner. New - 6.0 mm, Cond.- 1.0 mm	5.0	5.0
13	The diameter of retracing/locking pin & condition Diameter- 16.0 mm	15.00	15.00
14	Hand brake	New modified H/Brake provided, MECHWELL, SL No-13529	

CBC - Bhavin, Gyano Ram
H/Brake - E. Khan, Diru

Name of Staff

Signature of SE/JE

Signature of SSE